

Abhinav Hiroji

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SUMMARY

Aspiring Data Scientist and Software Developer with expertise in Python, C++, SQL, and web technologies. Experienced in building machine learning models, developing scalable software, and deploying full-stack applications. Proficient in time series forecasting, data analysis, deep learning (LSTM, TensorFlow, Keras), and backend development (Flask, REST APIs). Passionate about applying both data-driven insights and engineering skills to create impactful, real-world solutions.

EDUCATION

10/2022 – 07/2026 **Jain University Faculty of Engineering and Technology**
Bangalore, Karnataka

TECHNICAL SKILLS

Languages

Python, C++, SQL, HTML, CSS, JavaScript

Database

MYSQL

Concepts

Machine Learning, Deep Learning, Time Series
Analysis, Data Preprocessing, Model Evaluation, Data
Visualization, OOP, Data Structures & Algorithms

Libraries/Frameworks

Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn,
TensorFlow, Keras, Flask, Streamlit

Tools

PowerBI, Excel, Jupyter Notebook

PROJECTS

Air Quality Index (AQI) Prediction using LSTM

- Developed a machine learning-based web application to predict AQI using multivariate time series data (PM2.5, PM10, CO, NO2, etc.).
- Trained an LSTM deep learning model (TensorFlow/Keras) achieving strong prediction accuracy
- Deployed the Gemini API to ingest real-time air quality data, creating a robust data pipeline for analysis and reporting, which supported 10+ research initiatives.
- Deployed interactive dashboard for AQI category visualization and health insights

Smart Home Automation using Arduino

- Designed an Arduino-based automation system to control home appliances like lights, fans, and security alarms using sensors and IoT modules.
- Enable automation based on temperature, light intensity, and motion.
- Utilized components like LDR, IR sensors, relays, and optional ESP8266 for wireless control.
- Focused on low-cost, scalable automation suitable for energy-efficient smart homes.

Coursework

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| • Operating Systems | • Database Management System |
| • Machine Learning | • Computer Networks |
| • Data Structures & Algorithms | • Artificial Intelligence |
| • Probability & Statistics | |